EE2021: Tutorial - 6 (Oct 14th, 2025)

1. Show that $n(E) = g_C(E)f(E)$ peaks at $E_C+kT/2$ and similarly $p(E)=g_V(E)(1-f(E))$ peaks at $E_V-kT/2$. Assume the semiconductor is non-degenerate and Maxwell-Boltzmann approximation can be used.

2. Consider a Silicon sample with $N_A = 10^{15}cm^{-3}$ at 300 K. Compute and plot the equilibrium carrier concentrations (electron concentration in left Y axis and hole in right Y axis) vs N_D in a log-log plot for different values of $N_D = \{10^{13}, 10^{14}, 10^{15}, 10^{16}, 10^{17}\}$. Plot Fermi level with respect to the middle of the bandgap (E_F-E_i) vs N_D in a semi-log plot.

3. A Silicon device is doped with donor impurity atoms at a concentration of 10^{15} cm^{-3}. For the device to operate properly, the intrinsic carriers must contribute no more than 5 percent to the total electron concentration. (a) What is the maximum temperature that the device may operate? (b) What is the change in position of E_F from the T =300 K value to the maximum temperature value determined in part (a)

4. Partially ionized donor concentration at a low temperature can be written as

$$N_D^+ = N_D - N_D^o = \frac{N_D}{1 + \beta_d \exp\dfrac{E_F - E_D}{kT}}$$

where E_D is donor level, $E_D=E_C-E_d$ and E_d is the donor ionization energy and β_d is donor degeneracy factor which is typically 2. Show that for low enough temperature, $kT << E_d$, electron concentration can be written as

$$n_o \simeq \sqrt{\frac{N_D N_c}{\beta_d}} \exp\left(-\frac{E_d}{2kT}\right)$$

<u>Food for thought</u>

5. In an extrinsic semiconductor, why is $n \neq N_D+n_i$ and $p \neq n_i$ even though charge neutrality is maintained as $p - n + N_D^+=0$?

6. For high temperatures, Maxwell–Boltzmann statistics might not be suitable for an intrinsic semiconductor. How can this happen?

7. Describe a semiconductor in which Maxwell–Boltzmann statistics cannot be used in thermal equilibrium for neither electrons nor holes at room temperature. Explain

8. What could be the location of Fermi level for an extrinsic semiconductor at 0 K for these cases a) n-type with only donors b) p-type with only acceptors

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$n[E] = g_c(E) f(E)$ $g_c[E] \rightarrow$ density of states in CB

$f(E) \rightarrow$ fermi dirac occupational probability

$g_c(E) \propto \sqrt{E - E_c}$

Maxwell boltzman approximation: $f(E) \approx e^{-(E-E_f)/K_BT}$

$n[E] = A\sqrt{E-E_c}\, e^{-E/K_BT}$

$E_f \rightarrow$ fermi level, probab of finding $e^{\ominus}$ is 50% exact

At 0K: highest energy level occupied by $e^{\ominus}$s

$$f(E) = \frac{1}{1 + e^{(E-E_f)/K_BT}}$$

$$E - E_c = \frac{\hbar^2 k^2}{2m^*} \Rightarrow k = \sqrt{\frac{2m^*(E-E_c)}{\hbar^2}}$$

$$N(k) = \frac{4/3\pi k^3}{(2\pi/L)^3} \times 2 = \frac{k^3}{3\pi^2}$$

$$N(E) = \frac{1}{3\pi^2}\left(\frac{2m^*(E-E_c)}{\hbar^2}\right)^{3/2}$$

$$g_c(E) = \frac{dN(E)}{dE}$$

$$\frac{dN(E)}{dE} = 0 \Rightarrow A\sqrt{E-E_c}\, e^{-E/K_BT}\left(\frac{-1}{K_BT}\right) + Ae^{-E/K_BT}\frac{1}{2\sqrt{E-E_c}} = 0$$

$$\Rightarrow 2(E-E_c) = K_BT \rightarrow E = E_c + K_BT/2$$

$p(E) \rightarrow$ hole distribution $g_v(E) \rightarrow$ DOS in VB

$g_v(E) \propto \sqrt{E_v - E}$ $1 - f(E) \sim e^{-(E_f - E)/K_BT}$

$$\Rightarrow p(E) = B\sqrt{E_v - E}\, e^{E/K_BT}$$

$$\frac{dp(E)}{dE} = 0 \rightarrow E = E_v - K_BT/2$$

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Silicon sample with acceptors 10^{15}/cc → p-type full of holes

$10^{13} \Rightarrow p_0 \approx N_A - N_D \approx N_A = 10^{15}$/cc

$n_i = 10^{10} \Rightarrow 10^{20}/10^{15} = 10^5$/cc $= n_0$

$10^{14} \Rightarrow p_0 = 9 \times 10^{14}$/cc $\Rightarrow n_0 = 1.1 \times 10^5$ /cc

$10^{15} \Rightarrow p_0 \approx n_i = n_0 = 10^{10}$/cc

$10^{16} \Rightarrow n_0 \approx N_D - N_A = 9 \times 10^{15} \Rightarrow p_0 = 1.1 \times 10^4$/cc

$10^{17} \Rightarrow n_0 \approx N_D - N_A \approx 10^{17} \Rightarrow p_0 = 10^3$/cc

$$E_f - E_i = K_B T \ln(n_0/n_i)$$

$$E_i = E_{midgap} + \frac{3}{4} K_B T \ln\left(\frac{m_h^*}{m_e^*}\right)$$

$E_{midgap} \rightarrow \frac{E_c + E_v}{2}$

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$n = N_d + p \qquad p = 0.05n$

$\Rightarrow 0.95n = N_d \qquad n_i^2 = np \Rightarrow n_i = 2.35 \times 10^{14}$

$n_i(T) = 5.23 \times 10^{15} T^{3/2} e^{-E_g/2KT} \qquad E_g = 1.1 \Rightarrow T_{max} = 508$

$E_f - E_{fi} = KT \ln(n/n_i) \qquad KT_{max} = 0.287\ eV$

$$N_D^+ = N_D - N_D^o = \frac{N_D}{1 + \beta_d \exp \frac{E_F - E_D}{kT}}$$

$n \approx N_D^+$ $\quad n = \dfrac{N_D}{1 + \frac{2}{N_C} e^{\frac{E_F - E_C}{kT}} \cdot e^{\frac{E_C - E_D}{kT}}}$ $\quad n = \dfrac{N_D}{1 + \frac{2}{N_C} \cdot n \cdot e^{(E_C - E_D)/kT}}$

$x N_C$

$$n = \frac{-1 + \sqrt{1 + \frac{8 N_D e^{E_d/kT}}{N_C}}}{\frac{4 e^{E_d/kT}}{N_C}} = \frac{N_C}{4} e^{-E_d/kT} \sqrt{\frac{8 N_D}{N_C}} e^{-E_d/2kT}$$